

Final Project - Computational Mathematics

Variant 1

Instructions

Implement the assigned tasks, ensuring that they include:

- Source code with comments explaining all important steps.
- Screenshots of program outputs and explanations.
- Graphs or tables where applicable.
- Detailed explanations for your approaches and results.

For the final project you will be required to develop an integrated interface that allows for user interaction with various computational tasks. The interface should enable users:

- to change methods dynamically (e.g., droplist or radio buttons with all the methods within your variant)
- to input coefficients for equations
- enter values into matrices
- provide intervals, initial guesses/approximations, and etc.
- specify integrals
- enter subinterval values
- manipulate arrays (data)
- add additional parameters depending on the method (e.g., omega, accuracy value, and etc.).

Additionally, it should provide buttons for:

- executing calculations
- plotting graphs/tables if it's applicable.

And a designated area for displaying results.

Save your work as a PDF report and submit it to Moodle before the deadline.

P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=x^3-4x+1$ in the range $x \in [0,3]$. Use the graph to find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=x^2-5$ in the interval $[2,3]$ using:

- Bisection method.
 - Secant method.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Jacobi Method

Problem: Solve the system of equations:

$$x+y+z=6$$

$$2y+5z=-4$$

$$2x+3y+z=27$$

Start with an initial guess $x_0=[0,0,0]$.

Task 4: Iterative Method for Matrix Inversion

Problem: Compute the inverse of matrix

$$A = \begin{pmatrix} 4 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 4 \end{pmatrix}$$

using an iterative method with an initial guess based on its trace.

Task 5: Linear Curve Fitting

Problem: Fit a line to data points:

$(1,2),(2,3),(3,5),(4,7),(5,11)$ using least squares.

Task 6: Newton's Forward Interpolation Formula

Problem: Given data points $x=[0,1,2,3]$ and $y=[1,4,9,16]$, estimate $f(1.5)$.

Task 7: First Derivative Using Newton's Forward Difference Formula

Problem: Given data points $x=[0,1,2]$ and $y=[1,8,27]$, estimate dy/dx at $x=1$.

Task 8: Trapezoidal Rule

Problem: Approximate $\int_0^1 (x^2+x)dx$ using the Trapezoidal Rule with 4 subintervals. Compare with the exact value.

Final Project - Computational Mathematics

Variant 2

Instructions

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Additionally, it should provide buttons for:

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And a designated area for displaying results.

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Task 1: Graphical Method and Absolute Error

Problem: Plot $f(x)=e-x-x^2$ over $[0,2]$ to find an approximate root and calculate absolute error against a numerical solution.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=x^3-x-2$ in the interval $[1,2]$ using:

- Bisection method.
 - Secant method.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Gauss-Seidel Method

Problem: Solve the system of equations:

$$x+y+z=9,$$

$$x+z=3,$$

$$y+z=7$$

Start with an initial guess $x_0=[0,0,0]$.

Task 4: LU Factorization

Problem: Perform LU factorization on matrix

$$A = \begin{pmatrix} 3 & -2 & 5 \\ -4 & 6 & 7 \\ -5 & 8 & 9 \end{pmatrix}$$

Task 5: Polynomial Curve Fitting

Problem: Fit a quadratic curve to data points: $(0,0),(1,1),(2,4),(3,9),(4,16)$.

Task 6: Lagrange's Interpolation Formula

Problem: Estimate $f(5.5)$ given points: $(5,25),(6,36),(7,49),(8,64)$.

Task 7: Second Derivative Using Newton's Forward Difference Formula

Problem: Given data points $x=[5,6,7,8]$ and $y=[25,36,49,64]$, estimate d^2y/dx^2 at $x=1$.

Task 8: Simpson's Rule

Problem: Estimate $\int_0^\pi \sin(x)dx$ using Simpson's rule with 6 subintervals.

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Variant 3

Instructions

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- enter subinterval values
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Additionally, it should provide buttons for:

- executing calculations
- plotting graphs/tables if it's applicable.

And a designated area for displaying results.

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P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=x^4-10x^2+9$ in the range $x \in [-4,4]$. Use the graph to find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=x^3-6x^2+11x-6$ in the interval $[0,3]$ using:

- Bisection method.
 - Newton-Raphson methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Relaxation Method

Problem: Solve the system of equations using relaxation with parameter $\omega=0.8$:

$$x+y+z=10$$

$$x+z=6$$

$$y+z=8$$

Task 4: Finding Eigenvalues Using Power Method

Problem: Use Power Method to find largest eigenvalue of matrix:

$$A = \begin{pmatrix} 6 & 2 & 3 \\ 2 & 6 & 4 \\ 3 & 4 & 6 \end{pmatrix}$$

Task 5: Exponential Curve Fitting

Problem: Fit model to data points:

$$(0,1), (1,e), (2,e^2), (3,e^3).$$

Task 6: Cubic Spline Interpolation

Problem: Use cubic spline interpolation on data points: $(0,0), (0.5,0.25), (1.0,0.75), (1.5,2.25)$.

Task 7: Picard's Method

Problem: Solve the differential equation $dy/dx=x+y$, with initial condition $y(0)=1$, up to the fourth approximation. Find the value of y at $x=0.2$.

Task 8: Simpson's 1/3 Rule

Problem: Using Simpson's 1/3 Rule, compute $\int_0^\pi \sin(x) dx$ with 10 subintervals. Verify the accuracy by comparing it to the exact integral value.

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Variant 4

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Additionally, it should provide buttons for:

- executing calculations
- plotting graphs/tables if it's applicable.

And a designated area for displaying results.

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P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=\cos(x)-x$ in the range $x\in[0,1]$. Use the graph to find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=x^2-4\sin(x)$ in the interval $[0,3]$ using:

- False position methods.
 - Newton-Raphson methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Jacobi Method

Problem: Solve the system of equations:

$$3x+y-z=1$$

$$2x-8y+z=-2$$

$$-x+y+5z=3$$

Start with an initial guess $x_0=[0,0,0]$

Task 4: Iterative Method for Matrix Inversion

Problem: Compute the inverse of matrix

$$A = \begin{pmatrix} 5 & -3 & 2 \\ -3 & 9 & -1 \\ 2 & -1 & 7 \end{pmatrix}$$

using an iterative method with an initial guess based on its trace.

Task 5: Linear Curve Fitting

Problem: Fit a line to data points:

$(1,5),(2,8),(3,12),(4,15),(5,20)$ using least squares.

Task 6: First Derivative Using Newton's Forward Difference Formula

Problem: Given data points $x=[0,1,2]$ and $y=[1,3,7]$, estimate dy/dx at $x=1$.

Task 7: Taylor Series Method

Problem: Using the Taylor series method, solve $dy/dx=y^2+x^2$, $y(0)=1$, to compute $y(0.1)$ and $y(0.2)$. Use up to the third derivative for approximation.

Task 8: Simpson's 3/8 Rule

Problem: Using Simpson's 3/8 Rule, compute $\int_2^5 x^3 dx$ with 6 subintervals. Provide the absolute error compared to the exact value.

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Variant 5

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- executing calculations
- plotting graphs/tables if it's applicable.

And a designated area for displaying results.

Save your work as a PDF report and submit it to Moodle before the deadline.

P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=x^4-x^3+x^2-x+1$ in the range $x\in[-2,2]$. Find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=e-x+x^2-3$ in the interval $[0,3]$ using:

- Iteration methods.
 - Newton-Raphson methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Gauss-Seidel Method

Problem: Solve the system of equations:

$$4x+y+z=12$$

$$x+5y+z=15$$

$$x+y+6z=10$$

Start with an initial guess $x_0=[0,0,0]$.

Task 4: LU Factorization

Problem: Perform LU factorization on matrix

$$A = \begin{pmatrix} 10 & -7 & 0 \\ -3 & 6 & -2 \\ 0 & -3 & 9 \end{pmatrix}$$

Task 5: Polynomial Curve Fitting

Problem: Fit a cubic curve to data points $(0,0),(1,1),(2,8),(3,27)$.

Task 6: Lagrange's Interpolation Formula

Problem: Estimate $f(7)$ given points: $(6,36),(8,64),(10,100),(12,144)$.

Task 7: Euler's Method

Problem: Solve $dy/dx=x-y$, $y(0)=2$, using Euler's method with a step size $h=0.1$, for $0\leq x\leq 0.5$. Provide results for each step.

Task 8: Boole's Rule

Problem: Calculate $\int_1^3 e^{-x} dx$ using Boole's Rule. Use 8 subintervals and compare the result with the exact integral value.

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Variant 6

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Additionally, it should provide buttons for:

- executing calculations
- plotting graphs/tables if it's applicable.

And a designated area for displaying results.

Save your work as a PDF report and submit it to Moodle before the deadline.

P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=x^5-4x^4+6x^3-4x+1$ in the range $x \in [0,5]$. Find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=\ln(x)-x/10$ in the interval $[0.01, 10]$ using:

- False position methods.
 - Newton-Raphson methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Relaxation Method

Problem: Solve the system of equations using relaxation method with parameter $\omega=0.9$:

$$x+y+z=9$$

$$2x-3y+4z=13$$

$$3x+4y+5z=40$$

Task 4: Finding Eigenvalues Using Power Method

Problem: Use Power Method to find largest eigenvalue of matrix

$$A = \begin{pmatrix} 8 & 4 & 2 \\ 4 & 8 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Task 5: Exponential Curve Fitting

Problem: Fit model to data points $(0.5,2),(1.5,6),(2.5,18),(3.5,54)$.

Task 6: Cubic Spline Interpolation

Problem: Use cubic spline interpolation on data points: $(0.5,0.25),(1.5,0.75),(2.5,2.25),(3.5,6.25)$ to estimate values at specified locations.

Task 7: Modified Euler's Method

Problem: Using the modified Euler method, compute $y(0.4)$ for $dy/dx=\sin(x)-y$, $y(0)=1$, with step size $h=0.2$.

Task 8: Weddle's Rule

Problem: Calculate $\int_0^6 1/(1+x^2) dx$ using Weddle's Rule. Divide the interval into 6 subintervals. Discuss how the rule handles this function.

Final Project - Computational Mathematics

Variant 7

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Save your work as a PDF report and submit it to Moodle before the deadline.

P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=x^3-3x+2$ in the range $x \in [-2,2]$. Find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=x^4-5x^2+4$ in the interval $[0,3]$ using:

- False position methods.
 - Bisection methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Gaussian Elimination with Partial Pivoting

Problem: Solve the system using Gaussian elimination with partial pivoting:

$$\begin{aligned} 2x+y-z &= 8 \\ -3x+y+2z &= -11 \\ -2x+y+3z &= -3 \end{aligned}$$

Task 4: Iterative Method for Matrix Inversion

Problem: Compute the inverse of matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 5 & 6 & -1 \end{pmatrix}$$

using an iterative method with an initial guess based on its trace.

Task 5: Polynomial Curve Fitting

Problem: Fit a quadratic curve to data points $(0,0),(1,1),(2,4),(3,9),(4,16)$ using least squares.

Task 6: Lagrange's Interpolation Formula

Problem: Given data points $x=[1,3,5]$ and $y=[2,8,18]$, estimate $f(4)$ using Lagrange's interpolation.

Task 7: Romberg's Integration

Problem: Apply Romberg's Integration to approximate $\int_0^1 x^2 dx$. Use $h=0.5, 0.25$, and 0.125 to demonstrate convergence. Show the Romberg Table and explain your results.

Task 8: Runge-Kutta 2nd Order

Problem: Apply the Runge-Kutta second-order method to solve $dy/dx=e^x-y$, $y(0)=0$, for $x=0.2$, with a step size $h=0.1$.

Final Project - Computational Mathematics

Variant 8

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And a designated area for displaying results.

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P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=e^{-x}-x^3+x$ in the range $x \in [0,3]$. Find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=x^5-x^4+x^3-x+1$ in the interval $[-2,2]$ using:

- Secant methods.
 - Bisection methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Jacobi Method

Problem: Solve the system using Jacobi method starting with initial guess of zeros:

$$x+y+z=6$$

$$2y+z=-4$$

$$x+y+z=-3$$

Task 4: LU Factorization

Problem: Perform LU factorization on matrix

$$A = \begin{pmatrix} 4 & -2 & -1 \\ -2 & 4 & -2 \\ -1 & -2 & 4 \end{pmatrix}$$

Task 5: Polynomial Curve Fitting

Problem: Fit a quadratic curve to data points $(1,1),(5,25),(6,36),(7,49),(8,64)$ using least squares.

Task 6: Cubic Spline Interpolation

Problem: Use cubic spline interpolation on data points: $(0.5,0.25),(1.5,0.75),(2.5,2.25),(3.5,6.25)$ to estimate values at specified locations.

Task 7: Trapezoidal Rule

Problem: Approximate the integral $\int_4^1 (\ln(x)/x) dx$ using the Trapezoidal Rule with 6 subintervals. Compare the approximate result with the actual value of the integral.

Task 8: Runge-Kutta 4th Order

Problem: Solve $dy/dx=x^2-y$, $y(0)=1$, using the Runge-Kutta 4th-order method for $0 \leq x \leq 0.3$ with $h=0.1$. Show the results at each step.

Final Project - Computational Mathematics

Variant 9

Instructions

Implement the assigned tasks, ensuring that they include:

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And a designated area for displaying results.

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P.S. Please be prepared to explain your code/solution/answers.

Task 1: Graphical Method and Absolute Error

Problem: Plot the graph of $f(x)=x^6-x^5+x^4-x+e^{-x}$ in the range $x\in[0,5]$. Find an approximate root and calculate the absolute error compared to the root found using a numerical method.

Task 2: Comparison of Root-Finding Methods

Problem: Find the root of $f(x)=\tan(x)-x$ in the interval $[0, \pi/4]$ using:

- Secant methods.
 - Newton-Raphson methods.
 - Measure iterations and calculate relative errors for both methods.
-

Task 3: Gauss-Seidel Method

Problem: Solve the system using Gauss-Seidel method starting with initial guess of zeros:

$$4x+y+z=12$$

$$-x+y+z=5$$

$$-x+y+z=-7$$

Task 4: Finding Eigenvalues Using Power Method

Problem: Use Power Method to find largest eigenvalue of matrix:

$$A = \begin{pmatrix} 10 & 7 & 9 \\ 7 & 10 & 6 \\ 9 & 6 & 10 \end{pmatrix}$$

Task 5: Polynomial Curve Fitting

Problem: Fit a cubic curve to data points $(0,0),(1,0),(2,-8),(3,-27)$ using least squares.

Task 6: Lagrange's Interpolation Formula

Problem: Estimate $f(8)$ given points: $(6,36),(7,49),(9,81),(10,100)$.

Task 7: Error Analysis

Problem: Compare the numerical solution obtained by the Euler and Runge-Kutta methods for $dy/dx=x^2-y$, $y(0)=1$, for $x=0.3$, with $h=0.1$. Discuss the error in both methods and its implications.

Task 8: Runge-Kutta 4th Order

Problem: The growth of a bacterial culture is modeled by $dy/dt=2y-0.5y^2$, with $y(0)=1$. Using the Runge-Kutta 4th-order method, estimate $y(2)$ with a step size $h=0.5$.